

CSA5112 - Cryptography and Network Security

16. SHA512 HASH ALGORITHM

AIM:

To implement the **SHA-512 hashing algorithm** in Python and generate the hash value of a given message.

ALGORITHM:

1. Start the program.
2. Read the message from the user.
3. Convert the message into bytes.
4. Apply the SHA-512 hash function.
5. Convert the hash into hexadecimal format.
6. Display the SHA-512 hash.
7. Stop.

PYTHON PROGRAM:

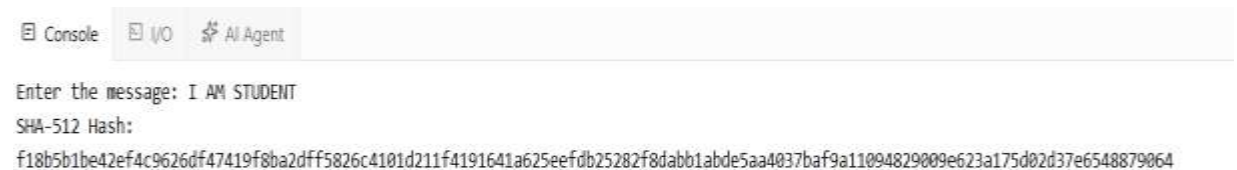
```
import hashlib

message = input("Enter the message: ")

sha512_hash = hashlib.sha512(message.encode())

print("SHA-512 Hash:", sha512_hash.hexdigest())
```

OUTPUT:



```
Enter the message: I AM STUDENT
SHA-512 Hash:
f18b5b1be42ef4c9626df47419f8ba2dff5826c4101d211f4191641a625eefdb25282f8dabb1abde5aa4037baf9a11094829009e623a175d02d37e6548879064
```